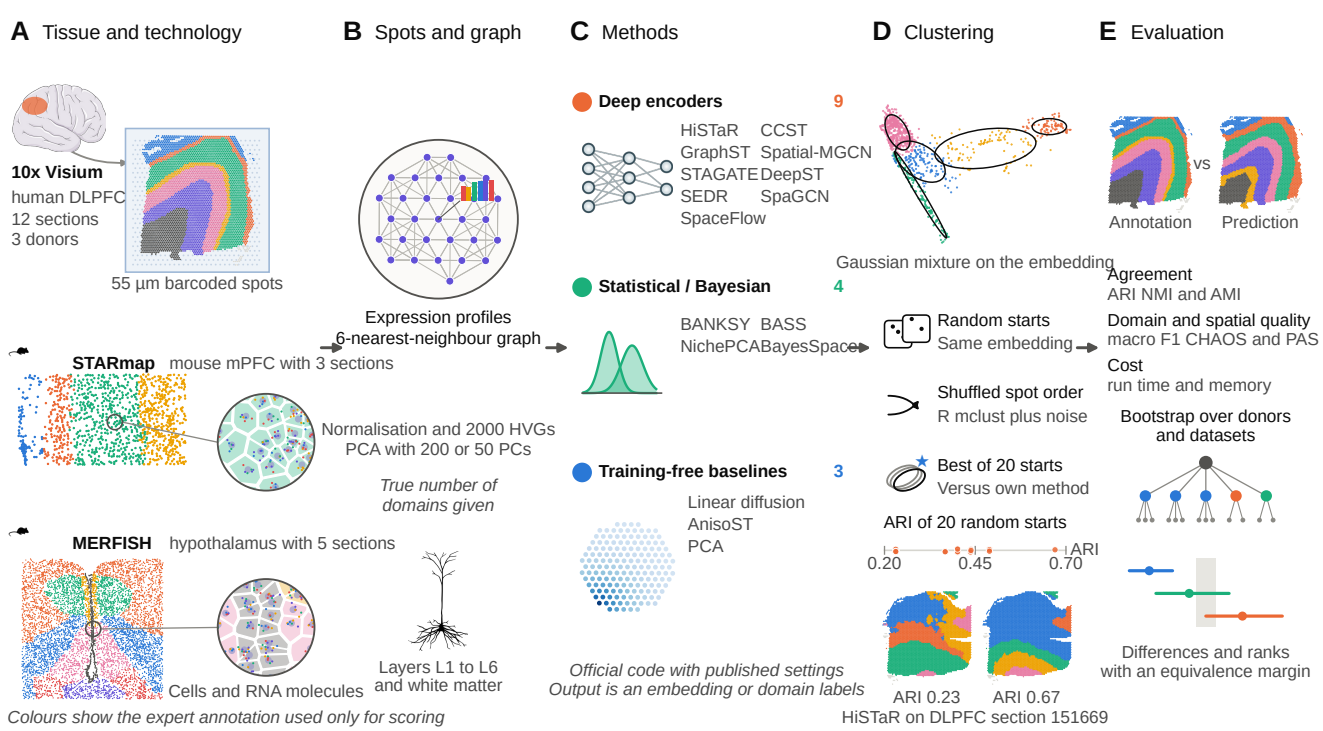

A Tissue and technology
B Spots and graph
C Methods
D Clustering
E Evaluation
10x Visium
human DLPFC
12 sections
3 donors
55 µm barcoded spots
Expression profiles
6-nearest-neighbour graph
STARmap mouse mPFC with 3 sections
Normalisation and 2000 HVGs
PCA with 200 or 50 PCs
True number of domains given
MERFISH hypothalamus with 5 sections
Cells and RNA molecules
Layers L1 to L6 and white matter
Colours show the expert annotation used only for scoring
Deep encoders 9
HiSTaR CCST
GraphST Spatial-MGCN
STAGATE DeepST
SEDR SpaGCN
SpaceFlow
Statistical / Bayesian 4
BANKSY BASS
NichePCA BayesSpace
Training-free baselines 3
Linear diffusion
AnisoST
PCA
Official code with published settings
Output is an embedding or domain labels
Gaussian mixture on the embedding
Random starts
Same embedding
Shuffled spot order
R mclust plus noise
Best of 20 starts
Versus own method
ARI of 20 random starts
0.20
0.45
0.70
ARI
ARI 0.23
ARI 0.67
HiSTaR on DLPFC section 151669
Annotation
vs
Prediction
Agreement
ARI NMI and AMI
Domain and spatial quality
macro F1 CHAOS and PAS
Cost
run time and memory
Bootstrap over donors and datasets
Differences and ranks with an equivalence margin